

5 Interesting Reference Designs FOR ROBOTICS

Looking to build robots without starting from scratch? These ready-made reference designs illustrate how to make robots work faster and smarter.

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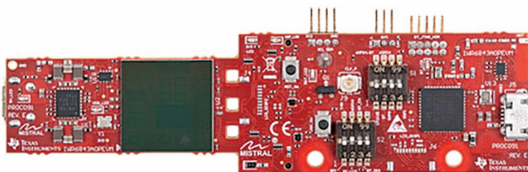
Robots are transforming how you work, live, and build systems. They take on tough, repetitive, or hazardous tasks, allowing you to focus on higher-value problem-solving. However, building a robot from the ground up is not easy. This is where reference designs prove valuable.

We have curated a few simple reference designs that address this challenge by serving as practical guides that show how motors, sensors, and controllers are integrated into working robotic systems.

Using these designs helps you reduce development time, test ideas more rapidly, and understand how real-world robots are engineered. Whether the focus is on radar-based navigation, high-performance motor control, or AI-driven decision-making, these setups support the step-by-step creation of more capable and reliable robots.

Autonomous robot

Autonomous robots are increasingly important across industries because they can operate continuously, handle repetitive or hazardous tasks, and deliver high accuracy. This improves safety, reduces costs, and saves time.



This reference design demonstrates an embedded robotic system that uses point cloud data from mmWave radar sensors and Robot Operating System (ROS) to process sensor inputs and control robot motion. By using mmWave sensors, the system becomes more compact, cost-effective, and faster to develop, enabling quicker testing and deployment of robotics applications.

The design supports applications such as factory sorting, inspection, and packaging; urban security and road monitoring; and drone-based obstacle avoidance and path planning. It integrates ROS on the processor, sensor interfaces, mobile robot control, and real-time visualisation on Ubuntu Linux. Using open source software makes it easy to update or extend the system with additional features.

Applications. This design supports the rapid development of robotics applications.

Link. <https://www.electronicsforu.com/electronics-projects/autonomous-robot-reference-design>

OEM Brand. Texas Instruments (TI)

Robotics and automation

This robotics and automation reference design is a complete kit demonstrating closed-loop



field-oriented control for brushless motors. It delivers high-precision motor control by combining an integrated control board with a high-performance motor and encoder.

The design includes current sensing, over-current protection, and an RS-485 interface, illustrating how motors, sensors, and controllers are integrated for reliable operation. The complete kit makes it well-suited for prototyping and for learning core robotics concepts without starting from scratch.

You can use this design to study